

(B) Training Pipeline

Stage 1: Supervised Fine-Tuning (SFT)

Function Prediction <i>"What is the function of this protein?"</i>	2.15M
Catalytic Activity <i>"Predict catalytic activity..."</i>	1.24M
General QA <i>"Describe this protein's properties..."</i>	0.99M
Domain Classification <i>"What domains does this contain?"</i>	0.46M

Total: 4.89M samples, ~2.1B tokens

Chat template | LoRA r=8 | 1 epoch



Stage 2: Group Relative Policy Optimization (GRPO)

GO F1 <i>"Predict Gene Ontology terms"</i> F1 score reward	10K
Stability ΔΔG <i>"Predict stability change"</i> Numerical accuracy reward	10K
Structure pLDDT <i>"Describe structure quality"</i> ESMFold pLDDT reward	10K

Total: 30K samples

Verifiable rewards | No human labels | KL penalty